



GOVERNMENT POLYTECHNIC AMRAVATI
(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

CURRICULUM DEVELOPMENT CELL

PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE: CM5465

COURSE TITLE: CLOUD COMPUTING

TEACHING SCHEME:

LEVEL OF COURSE	PRERE- QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
VI	CM3404	03	-	02	05	16	48	-	32

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HOURS	ESE		PA	TOTAL	ESE	PA	
3Hrs	MAX.	70	30*	100	25@	25^	150
	MIN.	28	---	40	10	10	

@: Internal Assessment #: External Assessment: Practical Based

(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Microproject-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1.RATIONALE:

Cloud computing has evolved as a very important computing model, which enables information, software, and other shared resources to be provisioned over the network as services in an on-demand manner. There are many aspects of cloud computing viz cloud types, storage in cloud, and security in cloud, cloud monitoring and management. Having specific skills in these areas is necessary for diploma pass-outs to create and maintain cloud based services. After learning this course student will be able to implement virtualization, create cloud based storage, Implement security, and manage cloud services.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

1. Maintain Cloud Based Application.
2. Implement virtualization in Cloud Computing.
3. Maintain Storage System in Cloud.
4. Maintain Cloud Services.
5. Implement Security in Cloud Computing.
6. Implement cloud on different platforms.

3.DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hr	Marks
Unit 1 Fundamental of Cloud Computing	1a.Explain the specified characteristics of Cloud Computing. 1b.Compare the given Cloud Deployment Models on the given criteria. 1c.Explain the given service offered by identified Cloud Service Model. 1d.Explain the given component of cloud computing architecture 1e. Write steps to use Cloud Based Integrated development Environment to develop the given application.	1.1 Cloud Computing, Essential characteristics of cloud computing 1.2 Cloud Deployment Model: Public cloud, Private cloud, Community cloud, Hybrid cloud 1.3 Cloud Service Models: IaaS, PaaS, SaaS 1.4 Cloud Economics and Benefits 1.5 Architecture of Cloud Computing 1.6 Cloud Computing Infrastructure 1.7 Cloud-Based Integrated Development Environment (IDE)	1	04	10
Unit 2 Virtualization	2a Explain the given feature of Virtualization. 2b.Explain the characteristics of the specified Virtualization type 2c. Write generic steps to build a virtual machine using VMware on the given OS. 2d.Describe the given disadvantage of Virtualization.	2.1 Introduction, Virtualization Reference Model, Characteristics of virtualized environment 2.2 Virtualization Types 2.3 Technology Example: VMware, Microsoft Hyper-V, KVM, Xen 2.4 Advantages: Virtual Machine (VM), VM Migration, VM Consolidation, VM Management 2.5 Disadvantages of Virtualization.	2	08	08
Unit 3 Storage in Clouds	3a.Explain the given component of storage system architecture. 3b.Write steps to design storage system for the given cloud set-up. 3c.Compare GFS and HDFS based on the given criteria.	3.1 Storage System Architecture, 3.2 Virtualized Data Centre (VDC) Architecture, VDC Environment, server, storage, networking, desktop and application virtualization techniques and benefits. 3.3 Block and file level storage virtualization, Virtual Provisioning, and automated storage tiering, Virtual storage area network (V SAN)	3	10	16

		and benefits, 3.4 Cloud file systems: Google File System GFS and Hadoop Distributed File System HDFS.			
Unit 4 Cloud monitoring and management	4a.Describe the given component of federated cloud computing. 4b.Compare different types of SLA based on the given criteria. 4c.Describe the given cloud interface standard. 4d.Explain the steps to use relevant technique for managing the given Cloud resource.	4.1 Service Provider and users 4.2 An architecture of federated cloud computing. 4.3 Service Level Agreement (SLA) management: Types of SLA, Life cycle of SLA. 4.4 Service catalog, management and functional interfaces of services , 4.5 Cloud portal and its functions 4.6 Cloud Service life cycle phases: Service planning, service creation, service operation and service termination 4.7 Cloud resource management Ab-initio Resource Assignment Periodic Resource Optimization	4	10	16
Unit 5 Security in Cloud Computing	5a.Explain the given security related risk in Cloud Computing. 5b.Explain the specified feature of Key security terminology for data security. 5c. Write steps to implement the given Technology for Securing the Data on cloud. 5d.Write steps to manage the Identity and Access facility of given Cloud set-up. 5e.Explain the given feature of Security-As-A-Cloud Service.	5.1 Cloud Security Fundamentals 5.2 Cloud Risk, Cloud Risk division Polity and Organizational Risks. Technical Risks Legal risks 5.3 Technologies for Data security, Data security risk 5.4 Digital identity and access management, 5.5 Content level security 5.6 Security-As-A-Cloud Service	5	08	10

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hr	Marks
Unit 6 Trends and future in cloud	6a.Explain the characteristics of the given Enabling Technology with the IoT. 6b. Select relevant cloud platform. 6c.Compare the feature of the given cloud platform on the given criteria.	6.1 Cloud trends in supporting Ubiquitous Computing 6.2 Enabling Technologies with Internet of Things (RI:11), SermorXS 6.3 Cloud platform such as Amazon EC2,S3,Microsoft Azure and Cloud stack,Google App Engine,Opensource cloud Euclyptus,Open Stack and Open Nebulla etc.	6	08	10

4.LIST OF PRACTICALS:

Sr No.	PRACTICAL OUTCOMES (PrOs)	CO NO.
1.	Use Google Doc to make spreadsheet and notes	1
2.	Install configure cloud using JustCloud	1
3.	Create/Delete Virtual Machine using VMware (Private Cloud)	2
4.	Implement storage Service on cloud using OpenStack	3
5.	Use OpenStack for file management	3
6.	Monitor cloud using Nagios Tool	4
7.	Create and Host Simple Web Application on Microsoft Azure/Google cloud/Any cloud platform(Part-I)	4
8.	Create and Host Simple Web Application on Microsoft Azure/Google cloud/Any cloud platform (Part-II)	4
9.	Implement Identity Management and Access Management using OpenStack	5
10	Configure Servers using Microsoft Azureto secure it.	5
11	Design a small application based on IoT using Arduino or Raspberry pi (Part-I)	6
12	Design a small application based on IoT using Arduino or Raspberry pi (Part-II)	6

Note

- The entire above listed practical's need to be performed compulsorily, so that the students reach the 'Precision level of Dave's Psychomotor Domain Taxonomy.
- The Process and Product related skills associated with each practical outcome shall be assessed on basis of following performance indicators.

S. No.	Performance Indicators	Weightage in %
1.	Installation Process and Attentiveness of student in practical's	20
2.	Write appropriate code to generate desired output in web application	20
3.	Debug, Test and Execute the programs	20
4.	Presentation of Output	20

S. No.	Performance Indicators	Weightage in %
5.	Able to Answer to oral questions	10
6.	Submission of report in time	10
Total		100%

The above PrOs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

- Handle and Monitor Cloud using different Tools
- Experiments with different clouds available.
- Configuration of Server using clouds.
- Demonstrate working as a leader or a team member.
- Follow ethical practices.

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1st year
- 'Organising Level' in 2nd year and
- 'Characterising Level' in 3rd year

5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related co-curricular activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Prepare journal of practicals.
- Undertake micro-projects.

6. SUGGESTED INSTRUCTIONAL STRATEGIES

Following are suggested instructional strategies, which the teacher can adopt for the attainment of the various outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- About 10-15% of the topic/ subtopic/ content which is relatively simple and descriptive in nature may be given to the students for self learning, for self directed learning and assess the development of CO's through class room presentation.
- Guide student(s) in undertaking micro-projects.
- Demonstrate students thoroughly before they start doing the practice.
- Use different instructional strategies in classroom teaching.
- Observe continuously and monitor the performance of student in lab.

7. SUGGESTED MICRO-PROJECTS.

Only one micro-project is to be undertaken by a student in the beginning of the semester. S/he ought to submit it by the end of semester. Each micro-project should include two or more Cos. The micro-project could be industry application based, internet based, laboratory based, field based and survey based. In the all semesters, it could be group-based (5 to 6 Student) to build up skill and confidence in every student. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The student ought to submit micro-project by the end of the semester to develop the industry oriented COs. A suggested list is given below.

- Prepare the report on case study of Amazon Cloud Services.
- Prepare the report on case study of Google App Engine.
- Create infrastructure as service using OpenStack.
- Develop Personal Cloud using ownCloud and Raspberry Pi.

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRE

Sr No.	Equipment Name with Broad Specification	Practical No.
1	Hardware : Computer system (i3 - i5 preferable) (Any computer system with basic configuration)	All Practical
2	Operating system : Windows / Linux	
3	Apache Tomcat, Java, Python, Virtualization software	
4	Academic version of any public cloud (Google/AWS/Azure)	

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks		
			R Level	U Level	A Level
1	Fundamentals of Cloud Computing	10	02	04	04
2	Virtualization	08	02	02	04
3	Storages in Clouds	16	04	08	04
4	Cloud Monitoring and Management	16	04	04	08
5	Security in Cloud Computing	10	02	04	04
6	Trends and Future in Cloud Computing	10	02	04	04
Total		70	16	26	28

R= Remember, **U**= Understanding, **A**=Application and above (*Bloom's Revised taxonomy*)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of **UOs**. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

10. SUGGESTED LEARNING RESOURCES:

Sr.No.	Title of Book	Author	Publication
1.	Cloud Computing Principles and paradigms	Buyya Rajkumar, J. Broberg, A. Goscinski	A John Willey & Sons, Inc., Publication ISBN: 978-0-470-88799-8
2.	Cloud Computing	Sharma Rishabh	Wiley Publication ISBN: 978-81-265-5306-8

3.	Mastering Cloud Computing	Buyya Rajkumar, Vecchiola Christian, Selvi S Thamarai	McGraw Hill Publication ISBN:978-1-25-902995-0
4.	Cloud Computing	Singh Shainlendra	Oxford university Press ,ISBN:9780199477388

11. SOFTWARE/LEARNING WEBSITES.

- <http://nptel.ac.in/courses/106105167/1>
- <https://www.techopedia.com/definition/2/cloud-computing>
- <https://onlinelibrarywiley.com/doi/book/10.1002/9780470910105>
- <https://www.chinacloud.cn/upload/2011-07/11073107539898.pdf>

12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

SR. NO.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1	Mr. M. R. Torney	Lecturer in Computer Engineering	Government Polytechnic Amravati
2	Dr.P.R.Satav	Lecturer in Computer Engineering	Government Polytechnic Amravati

Govt. Polytechnic, Programme Board of Studies Computer Engineering has approved the above course curriculum on 30/12/2020 and is adopted for Computer Engineering Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGINEERING
GOVERNMENT POLYTECHNIC, AMRAVATI.

The General Board of Studies has approved the above course curriculum on 06/02/2021

The Governing Body has approved the above course curriculum on 13/08/2020